

1. Write a program to store 15 integer values in an ArrayList and remove all duplicate elements without using another collection.

```
import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {

        ArrayList<Integer> list = new ArrayList<>();

        list.add(10);
        list.add(20);
        list.add(30);
        list.add(10);
        list.add(40);
        list.add(20);
        list.add(50);
        list.add(60);
        list.add(30);
        list.add(70);
        list.add(80);
        list.add(90);
        list.add(100);
        list.add(80);
        list.add(50);

        for (int i = 0; i < list.size(); i++) {
            for (int j = i + 1; j < list.size(); j++) {
                if (list.get(i).equals(list.get(j))) {
                    list.remove(j);
                }
            }
        }
    }
}
```

```

        j--;
    }
}
}

System.out.println(list);
}
}

```

Output

```

Main.java
1  import java.util.ArrayList;
2
3  public class Main {
4      public static void main(String[] args) {
5
6          ArrayList<Integer> list = new ArrayList<>();
7
8          list.add(10);
9          list.add(20);
10         list.add(30);
11         list.add(10);
12         list.add(40);
13         list.add(20);
14         list.add(50);
15         list.add(60);
16         list.add(30);
17         list.add(70);
18         list.add(80);
19         list.add(90);
20         list.add(100);
21         list.add(80);
22         list.add(50);
23
24         for (int i = 0; i < list.size(); i++) {
25             for (int j = i + 1; j < list.size(); j++) {
26                 if (list.get(i).equals(list.get(j))) {
27                     list.remove(j);
28                     j--;
29                 }
30             }
31         }
32
33         System.out.println(list);
34     }
35 }

```

input

[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

...Program finished with exit code 0
Press ENTER to exit console.

2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

```

import java.util.LinkedList;

public class Main {

    public static void main(String[] args) {

        LinkedList<Integer> list = new LinkedList<>();
    }
}

```

```

list.add(101);

list.add(102);

list.add(103);

list.add(104);


System.out.println("Forward:");

for (int i : list)

    System.out.println(i);


System.out.println("Reverse:");

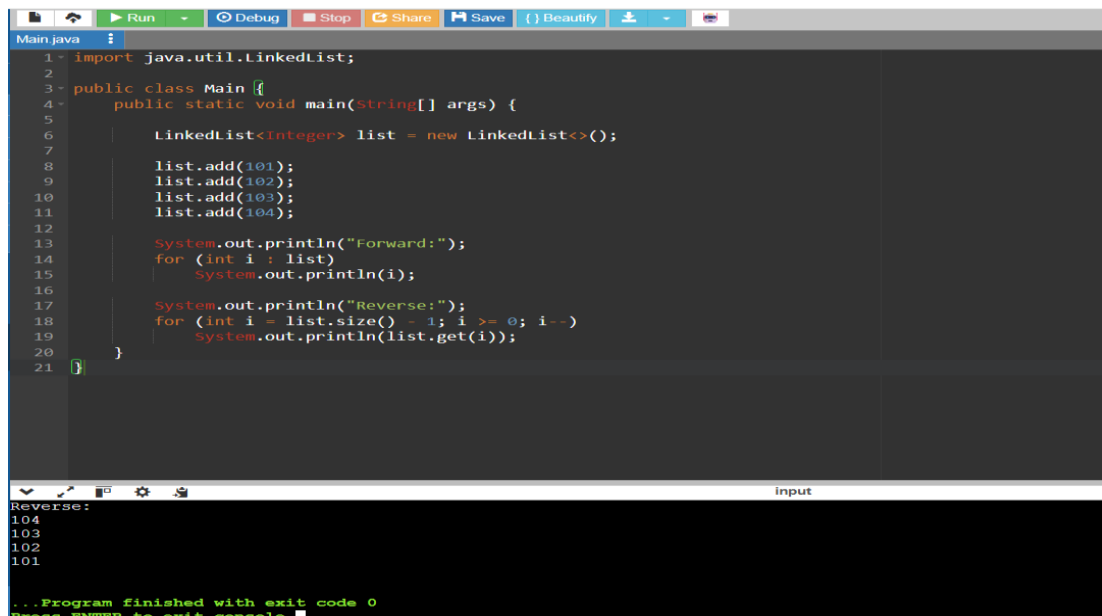
for (int i = list.size() - 1; i >= 0; i--)

    System.out.println(list.get(i));

}

}

```



The screenshot shows an IDE window titled 'Main.java' with the following code:

```

1 import java.util.LinkedList;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         LinkedList<Integer> list = new LinkedList<>();
7
8         list.add(101);
9         list.add(102);
10        list.add(103);
11        list.add(104);
12
13        System.out.println("Forward:");
14        for (int i : list)
15            System.out.println(i);
16
17        System.out.println("Reverse:");
18        for (int i = list.size() - 1; i >= 0; i--)
19            System.out.println(list.get(i));
20    }
21 }

```

The output console at the bottom shows the following text:

```

Reverse:
104
103
102
101
...Program finished with exit code 0
Press ENTER to exit console.

```

3. Store email IDs entered by the user in a HashSet. Display only unique email IDs and count the total unique entries.

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

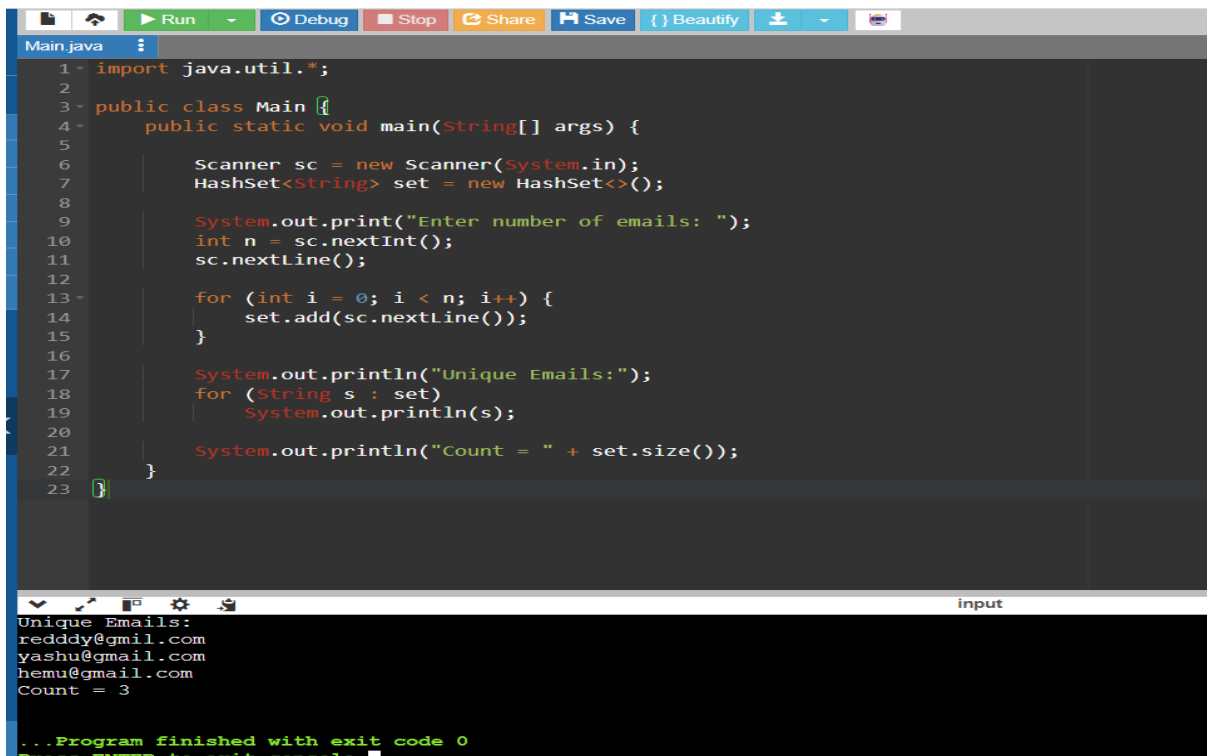
        Scanner sc = new Scanner(System.in);
        HashSet<String> set = new HashSet<>();

        System.out.print("Enter number of emails: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            set.add(sc.nextLine());
        }

        System.out.println("Unique Emails:");
        for (String s : set)
            System.out.println(s);

        System.out.println("Count = " + set.size());
    }
}
```



```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7         HashSet<String> set = new HashSet<>();
8
9         System.out.print("Enter number of emails: ");
10        int n = sc.nextInt();
11        sc.nextLine();
12
13        for (int i = 0; i < n; i++) {
14            set.add(sc.nextLine());
15        }
16
17        System.out.println("Unique Emails:");
18        for (String s : set)
19            System.out.println(s);
20
21        System.out.println("Count = " + set.size());
22    }
23 }
```

Unique Emails:
redddy@gmail.com
yashu@gmail.com
hemu@gmail.com
Count = 3

...Program finished with exit code 0
Press ENTER to exit console

4. Write a Java program to count the frequency of each word in a paragraph using a HashMap.

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

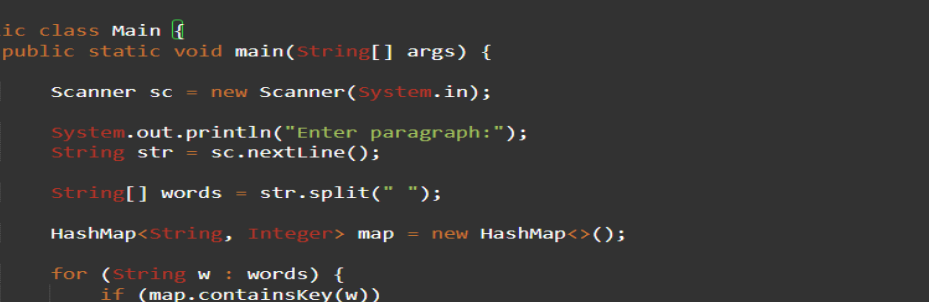
        System.out.println("Enter paragraph:");
        String str = sc.nextLine();

        String[] words = str.split(" ");

        HashMap<String, Integer> map = new HashMap<>();

        for (String w : words) {
            if (map.containsKey(w))
                map.put(w, map.get(w) + 1);
            else
                map.put(w, 1);
        }
    }
}
```

```
for (String key : map.keySet()) {
    System.out.println(key + " = " + map.get(key));
}
}
```



The screenshot shows an IDE with a Java file named `Main.java`. The code defines a `Main` class with a `main` method that uses a `Scanner` to read a paragraph, splits it into words, and counts the frequency of each word using a `HashMap`. The IDE's interface includes a top toolbar with icons for Run, Debug, Stop, Share, Save, Beautify, and other actions. The output window at the bottom shows the program's execution, displaying the input string and the frequency of each word.

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.println("Enter paragraph:");
9         String str = sc.nextLine();
10
11         String[] words = str.split(" ");
12
13         HashMap<String, Integer> map = new HashMap<>();
14
15         for (String w : words) {
16             if (map.containsKey(w))
17                 map.put(w, map.get(w) + 1);
18             else
19                 map.put(w, 1);
20         }
21
22         for (String key : map.keySet()) {
23             System.out.println(key + " = " + map.get(key));
24         }
25     }
26 }
```

input

```
^?java is easy
= 1
is = 1
easy = 1
java = 1

...Program finished with exit code 0
Press ENTER to exit console
```